## 21-3b How Changes in Income Affect the Consumer’s Choices

Now that we have seen how the consumer makes a consumption decision, let’s examine how this decision responds to changes in the consumer’s income. To be specific, suppose that income increases. As we have discussed, an increase in income leads to a parallel outward shift in the budget constraint, as in Figure 8. Because the relative price of the two goods has not changed, the slope of the new budget constraint is the same as the slope of the initial budget constraint.

**Figure 8**

### An Increase in Income

When the consumer’s income rises, the budget constraint shifts outward. If both goods are normal goods, the consumer responds to the increase in income by buying more of both of them. Here the consumer buys more pizza and more Pepsi.

The expanded budget constraint allows the consumer to choose a more desirable combination of pizza and Pepsi and therefore reach a higher indifference curve. Given the shift in the budget constraint and the consumer’s preferences as represented by her indifference curves, the consumer’s optimum moves from the point labeled “initial optimum” to the point labeled “new optimum.”

Notice that, in Figure 8, the consumer chooses to consume more Pepsi *and* more pizza. The logic of the model does not require increased consumption of both goods in response to increased income, but this situation is the most common. As you may recall from Chapter 4, if a consumer wants more of a good when her income rises, economists call it a **normal good**. The indifference curves in Figure 8 are drawn under the assumption that both pizza and Pepsi are normal goods.

Figure 9 shows an example in which an increase in income induces the consumer to buy more pizza but less Pepsi. If a consumer buys less of a good when her income rises, economists call it an
**inferior good** (a good for which an increase in income reduces the quantity demanded) . Figure 9 is drawn under the assumption that pizza is a normal good and Pepsi is an inferior good.

**Figure 9**

### An Inferior Good

A good is inferior if the consumer buys less of it when her income rises. Here Pepsi is an inferior good: When the consumer's income increases and the budget constraint shifts outward, the consumer buys more pizza but less Pepsi.

Although most goods in the world are normal goods, some are inferior goods. An example is bus rides. As income increases, consumers are more likely to own cars or Uber and less likely to ride the bus. Bus rides, therefore, are an inferior good.

FYI

**Utility: An Alternative Way to Describe Preferences and Optimization**

We have used indifference curves to represent the consumer's preferences. Another common way to represent preferences is with the concept of *utility*. Utility is an abstract measure of the satisfaction or happiness that a consumer receives from a bundle of goods. Economists say that a consumer prefers one bundle of goods to another if it provides more utility than the other.

Indifference curves and utility are closely related. Because the consumer prefers points on higher indifference curves, bundles of goods on higher indifference curves provide higher utility. Because the consumer is equally happy with all points on the same indifference curve, all these bundles provide the same utility. You can think of an indifference curve as an "equal-utility" curve.

The *marginal utility* of any good is the increase in utility that the consumer gets from an additional unit of that good. Most goods are assumed to exhibit *diminishing marginal utility*: The more of the good the consumer already has, the lower the marginal utility provided by an extra unit of that good.

The marginal rate of substitution between two goods depends on their marginal utilities. For example, if the marginal utility of good X is twice the marginal utility of good Y, then a person would need $2$ units of good Y to compensate for losing $1$ unit of good X, and the $MRS$ equals $2$. More generally, the marginal rate of substitution (and thus the slope of the indifference curve) equals the marginal utility of one good divided by the marginal utility of the other good.

Utility analysis provides another way to describe consumer optimization. Recall that, at the consumer's optimum, the marginal rate of substitution equals the ratio of prices. That is,

$$MRS = P_{\mathrm{X}}/P_{\mathrm{Y}}.$$

Because the marginal rate of substitution equals the ratio of marginal utilities, we can write this condition for optimization as

$$MU_{\mathrm{X}}/MU_{\mathrm{Y}} = P_{\mathrm{X}}/P_{\mathrm{Y}}.$$

Now rearrange this expression so that it becomes

$$MU_{\mathrm{X}}/P_{\mathrm{X}} = MU_{\mathrm{Y}}/P_{\mathrm{Y}}.$$

This equation has a simple interpretation: At the optimum, the marginal utility per dollar spent on good X equals the marginal utility per dollar spent on good Y. If this equality did not hold, the consumer could increase her utility by spending less on the good that provided lower marginal utility per dollar and more on the good that provided higher marginal utility per dollar.

When economists discuss the theory of consumer choice, they sometimes express the theory using different words. One economist might say that the goal of the consumer is to maximize utility. Another economist might say that the goal of the consumer is to end up on the highest possible indifference curve. The first economist would conclude that at the consumer's optimum, the marginal utility per dollar is the same for all goods, whereas the second would describe the optimum as the point at which the indifference curve is tangent to the budget constraint. In essence, these are two ways of saying the same thing.